DESIGN BRIEF · OCT 2026 · OWNER KR0

METRONIKS

An isometric voxel mech brawler. You drive and shoot in real time, fire heavy weapons on a timing hit, and tear the city down for scrap that rebuilds your mech between matches.

Genre mech brawler, extraction | Platform PC, keyboard and mouse | Mode single player vs AI | Budget 54.6M tokens

1 EXECUTIVE SUMMARY

Metr0niks drops you into Overpass Sector 4, a slum under a six-lane highway, in a mech you built from salvaged parts. You fight five waves of enemy mechs, with bosses on waves 3 and 5. Clear wave 5 and you extract with every scrap cube you picked up. Get wrecked and you lose all of it. Back in the garage you spend scrap on parts, and every part changes how the mech moves, heats up and hits.

PLAYER FANTASY	WIN	LOSE
Pilot a scrapyard war machine that wins by pulling a highway down on its enemies.	Clear wave 5 of an excursion. You extract and bank the match's scrap.	Your mech's armor hits 0, or you leave early. All of the match's scrap is lost; owned parts are kept.
THIS SEMESTER SHIPS One map, Excursion and Gauntlet modes, three enemy types and a 12-part garage.		

Every part is a tradeoff

Each part changes weight, heat or power, so building your mech changes how it moves and fights.

The city is a weapon

Pillars, overpasses and vats are as deadly as your guns. A cheap mech can beat an upgraded one by dropping a road on it.

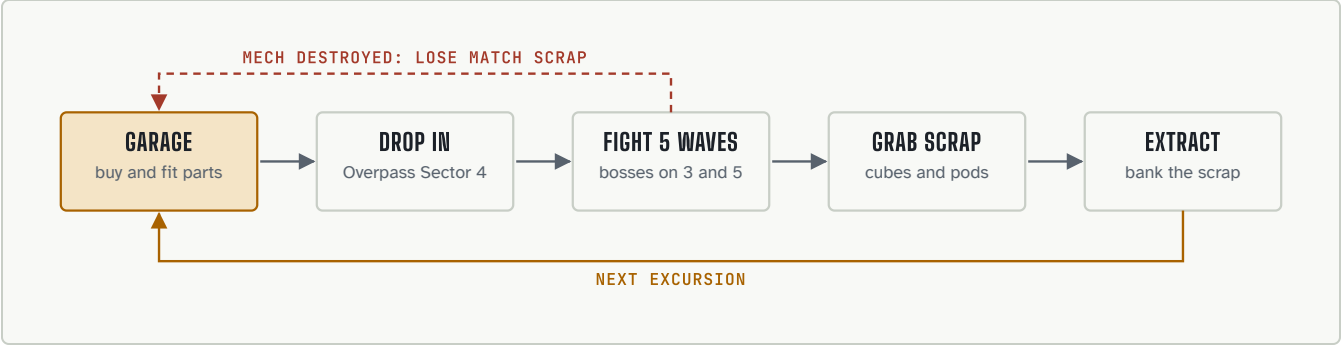
Real-time driving, timed heavy hits

Movement and light guns never pause. Heavy weapons reward a well-timed release.

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2 GAME MECHANICS

Core loop



One excursion. Dying at any point sends you back to the garage with none of that match's scrap.

1. **Build** in the garage: buy parts, fit them to six slots, and watch speed, armor and heat change before you confirm.
2. **Drop in** to Overpass Sector 4.
3. **Fight five waves.** Drive, shoot, time heavy hits, and break supports to drop buildings on enemies.
4. **Grab scrap and drop pods.** Destroyed enemies and vaults burst into scrap cubes you drive over to collect.
5. **Extract** after wave 5 to bank the scrap, then build again.

How you win and lose

WIN AN EXCURSION

Destroy every enemy in wave 5, including the boss. You extract automatically and the scrap you collected is banked.

LOSE AN EXCURSION

Your armor reaches 0 and the mech is destroyed, or you choose to leave before wave 5 ends. You go back to the garage with no scrap from that match. Parts you own are never lost, and a free starter mech means you can always deploy again.

Gauntlet runs the same waves with no penalty for losing and no progress for winning. It is where you practise a new build.

What you press

INPUT	WHAT HAPPENS ON SCREEN
WASD	The legs drive. Heavy builds drift and take wide turns; light builds snap around.
Mouse	The torso turns to the cursor, separately from the legs, so you can back away while firing forward. Aiming at upper floors tilts the guns up.
Left / right click	Fire the right or left arm. Light guns fire while held. A heavy gun with a full charge bar starts a shrinking timing ring when you hold, and fires when you release.
Q	Shoulder weapon or utility, such as the cryo vent's freezing blast.
Space	The legs' ability: a dash, a jump-jet burst onto a rooftop, or a siege anchor.

The heavy hit

Each heavy weapon has its own charge bar that fills while you fight. When it's full, hold the button: a ring shrinks toward a target ring over about one second, and you keep driving and aiming the whole time. Release on the target and the hit lands as a **Perfect**.

RELEASE	DAMAGE	WHAT YOU SEE
Perfect	200%	The shell cores straight through walls and takes out extra voxels, often the pillar behind your target.
Good	100%	A normal hit.

RELEASE	DAMAGE	WHAT YOU SEE
Miss, or the ring closes	50%	A weak hit that still costs full heat.

Every shot, jump and energy part adds heat. At 100% the mech overheats: weapons and thrusters shut off for 3 seconds and it vents scalding steam. An overheat during a ring cancels it as a Miss.

A minute in Overpass Sector 4

0:00	Wave 2. Three patrol bots come up the slum street under the highway. You back away with WASD while your torso stays on them, minigun chattering. The heat gauge climbs toward the red.
0:12	The bunker-buster's charge bar is full. You hold right click, the ring shrinks, and you release on the target: **Perfect**. The shell punches through the bots and cores the highway pillar behind them.
0:18	One more minigun burst and the pillar breaks. A dark shadow spreads across the road under the overpass. You hit Space and dash clear as 40 metres of road lands on the bots.
0:25	Scrap cubes pop up on the rubble. A drop-pod marker with a countdown appears two blocks away. A wounded heavy loader turns away from you and lumbers toward it.
0:40	You get there first, touch the pod and get Coolant flood: your heat locks at 0 for 10 seconds. You hold the minigun down on the loader until it bursts into double scrap.
0:55	The wave-3 boss, an ultra-light sniper, lands on a rooftop. You tilt your torso up to the top floor and start charging the mortar.

Building the mech

A mech has six slots: Core, right arm, left arm, shoulder, legs and reactor. The garage shows every change to power, armor, speed, weight and heat before you confirm. Mid-match, enough damage to an arm disables that weapon and drops its weight, so a battered mech gets faster; damaged legs slow you down. The starting pool is twelve parts, among them the minigun, shotgun, plasma cutter, bunker-buster cannon, pile-bunker spike, siege mortar, jump-jets and bulldozer treads.

3 AI ARCHITECTURE

MetrOniks has five in-game agents. Enemy mechs use the same parts, heat and damage rules as you, so you can read what they're about to do from the same cues you use on yourself. All of them run on fixed rules inside the game. None calls a language model while you play.

Patrol bot

WHAT YOU SEE

Light mechs walking routes through the streets. When you come into view they close to their gun's range and open fire.

EFFECT ON PLAY

The early-wave pressure and your main source of scrap. They overheat like you do, and an overheated bot breaks off and hides until it cools: your window to chase.

Heavy loader

WHAT YOU SEE

A slow, armored hauler carrying ore through the map.

EFFECT ON PLAY

A scrap piñata worth chasing. Shoot off its main weapon and it retreats out of sight, so you choose between finishing it and holding your ground.

Pod racer

WHAT YOU SEE

Any enemy breaks off the fight and runs for a drop pod when the pod is closer to it than to you, or when its own armor falls below half.

EFFECT ON PLAY

Every pod becomes a race. A wounded enemy fleeing toward a marker is telling you it's about to heal. Beat it there, or kill it on the way.

Demolisher

WHAT YOU SEE

When you park under a structure, enemies stop shooting you and shoot the pillar holding it up instead. A falling shadow appears on the ground before it lands.

EFFECT ON PLAY

The city works against you too. Standing still under the overpass gets punished, and the shadow gives you time to dash clear.

Bounty boss: the rooftop sniper

or jump-jets, or collapse the building it's standing on.

Wave director

EFFECT ON PLAY

Sets the rhythm of a match: fight, scramble for a pod, fight again. Gauntlet uses the same waves with the scrap penalty switched off.

How the agents play off each other. The demolisher and the pod racer pull in opposite directions. When a pod lands under the overpass, enemies race you to it, and the ones that lose the race start shooting the pillars above you. The rule you learn by wave 3 is to look up before you go for a pod.

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4 TECHNICAL STRATEGY

Development agents

Fourteen AI agents build Metr0niks. kr0 directs them and approves every design decision and every file they write.

AGENT	ROLE ON METR0NIKS
`creative-director`	Rejects any feature that doesn't make parts a tradeoff, the city a weapon, or heavy hits a timed release.
`technical-director`	Chose pre-split voxel chunks over full voxel physics, and signs off each architecture decision before code is written.
`producer`	Splits the semester into sprints, tracks tokens spent against the 54.6M budget, and moves stories below the cut line when a sprint overruns.
`game-designer`	Owns the GDD: run rules, the parts list, the timing-hit table and the drop pods.

AGENT	ROLE ON METR0NIKS
systems-designer	Writes the heat, charge-bar and weight formulas and keeps every tunable number in one table.
economy-designer	Sets scrap drops and garage prices so a cleared excursion pays for real upgrades and a death stings.
level-designer	Lays out Overpass Sector 4: which pillars hold the highway, where transports drive, and where pods can land.
gameplay-programmer	Builds the mech controls, weapons, timing ring, heat and drop pods.
engine-programmer	Builds the support graph that decides what falls when a pillar breaks, and the 300-piece debris cap.
ai-programmer	Builds the patrol, engage, pod-race and retreat behaviors, the pillar shooting and the sniper boss.
technical-artist	Turns the AI-made voxel models into game-ready parts and building chunks, and writes the cutaway and heat-glow shaders.
ux-designer	Designs the garage screen and the HUD: heat, charge bars, the timing ring and the weight change when an arm is lost.
qa-tester	Plays every finished story, keeps a screenshot as proof, and runs the 300-debris collapse test before any destruction change is merged.
prototyper	Built the firing-range prototype and builds throwaway tests for risks like timing the ring while driving.

How they hand work over. Combat features run as a small team: the game-designer writes the rule, the gameplay-programmer builds the player's side and the ai-programmer the enemies' side, the technical-artist adds the feedback, and the qa-tester closes the story only after watching it run. When two agents disagree, for example the engine-programmer's debris cap against the Perfect hit's bigger blast, the technical-director decides.

Semester scope

Assumes a 15-week semester ending in one playable vertical slice.

Ships this semester

- Overpass Sector 4 with collapsible highway and pillars
- Excursion and Gauntlet modes, 5 waves each
- Patrol bot, heavy loader and the rooftop sniper boss
- The 12-part starting pool in the first two rarity tiers
- Four drop pods, garage buying, fitting and repairs
- Saves stored on the player's machine

After the semester

- The Smelting Yards and Neon Financial Plaza maps
- Draft Duel and Scrap Rush multiplayer
- Rarity tiers blue, purple and orange
- Online accounts and server-side saves

Constraints

Collapse performance

A big collapse can drop the frame rate, and a stutter during a one-second timing ring turns a Perfect into a Miss. So buildings are pre-split into chunks, live debris is capped at 300 pieces with the oldest fading first, and a collapse test runs before any other feature is built.

No model calls during play

The timing ring lasts about a second and enemies react every frame. A call to a hosted model takes far longer than that and costs tokens on every match, so every in-game agent runs on fixed rules: zero runtime tokens per match.

One developer

kr0 is the only human, so everything tunable lives in data files the agents can change without code, and art is AI-made and approved by kr0.

API constraints on the agent crew

- **Context window.** The GDD and TDD are read one section at a time, so an agent building the heat system loads the heat section, not the parts list.
- **Rate limits.** At most three agents work in parallel, on stories that don't touch the same files.

- **Model tiers.** Directors run on the strongest model; specialists run on the session's model, which keeps most tokens on cheaper calls.

Token budget

Planning estimate for the semester build. About 85% of these tokens are input, because every agent call re-reads the GDD section and code it touches. The producer records actual spend per story in the tuning history and re-plans at each sprint close.

FEATURE	MAIN AGENTS	TOKENS
Design and architecture upkeep, stories	game-designer, technical-director, producer	3.0M
Voxel destruction and debris cap	engine-programmer, prototyper	6.0M
Mech model, slots, stats, disabled arms and legs	gameplay-programmer, systems-designer	4.0M
Light guns, heat, charge bars and timing ring	gameplay-programmer, systems-designer	5.0M
Camera, building cutaway, falling shadows	technical-artist, gameplay-programmer	3.5M
Enemy behaviors and sniper boss	ai-programmer	5.0M
Waves, drop pods, Excursion and Gauntlet rules	gameplay-programmer, level-designer	3.0M
Garage, scrap economy, local save	ux-designer, economy-designer	3.5M
Voxel art pipeline	technical-artist	2.5M
Playtests, screenshot evidence, regression runs	qa-tester	3.5M
Subtotal		39.0M
Rework reserve (40%)		15.6M
Total development budget		54.6M
Runtime tokens per match		0